

Nuclear Half-Life: Demonstration

Notes for teachers:

This is a demonstration of half-life and the random nature of radio-active decay and is intended support the medical physics module of the HSC.

Materials:

- At least 100 dice, with one 'marked' die to distinguish it from the others (could be a different colour, marked on each side with pen/paint).
- A container (e.g. a plastic tub) or a bag to tip the dice from.
- Graph paper for each student.

Activity Summary:

- The students will roll a large number of dice, where each die represents the nucleus of an atom.
- When a die roles a 1, that 'nucleus' can be said to have decayed and is removed from play.
- No matter how many dice the students have to begin with, after 3 rolls the number of dice should be (approximately) half – i.e. the dice have a half-life of 3 rolls.
- A time element could be introduced by having the students roll the dice at regular intervals, e.g. rolling the dice every 2 minutes would give a half-life of 6 minutes.
- By working in groups and averaging the results, this should also demonstrate the effect of sample size – 'better' decay curves will be shown by averaging the results.
- As each dice will 'decay' at random it is possible to predict when the marked dice (green in the picture below) will decay.



This activity works well in small groups with 100 dice each, however there are a many ways to run the demonstration and your choice will likely depend on the number of students (and dice available); two options are presented here, but of course you might choose a different configuration!

Option 1: is designed to be run as part of a rotation of activities, so that the class is split into small groups with only one group working on this at a time.

Option 2: is designed for the whole class to work in parallel. But with the pupils split into smaller groups with a set of dice each - not necessarily 100 dice for each group, but at least 20 is ideal.

Option 1: Small groups, 100 dice:

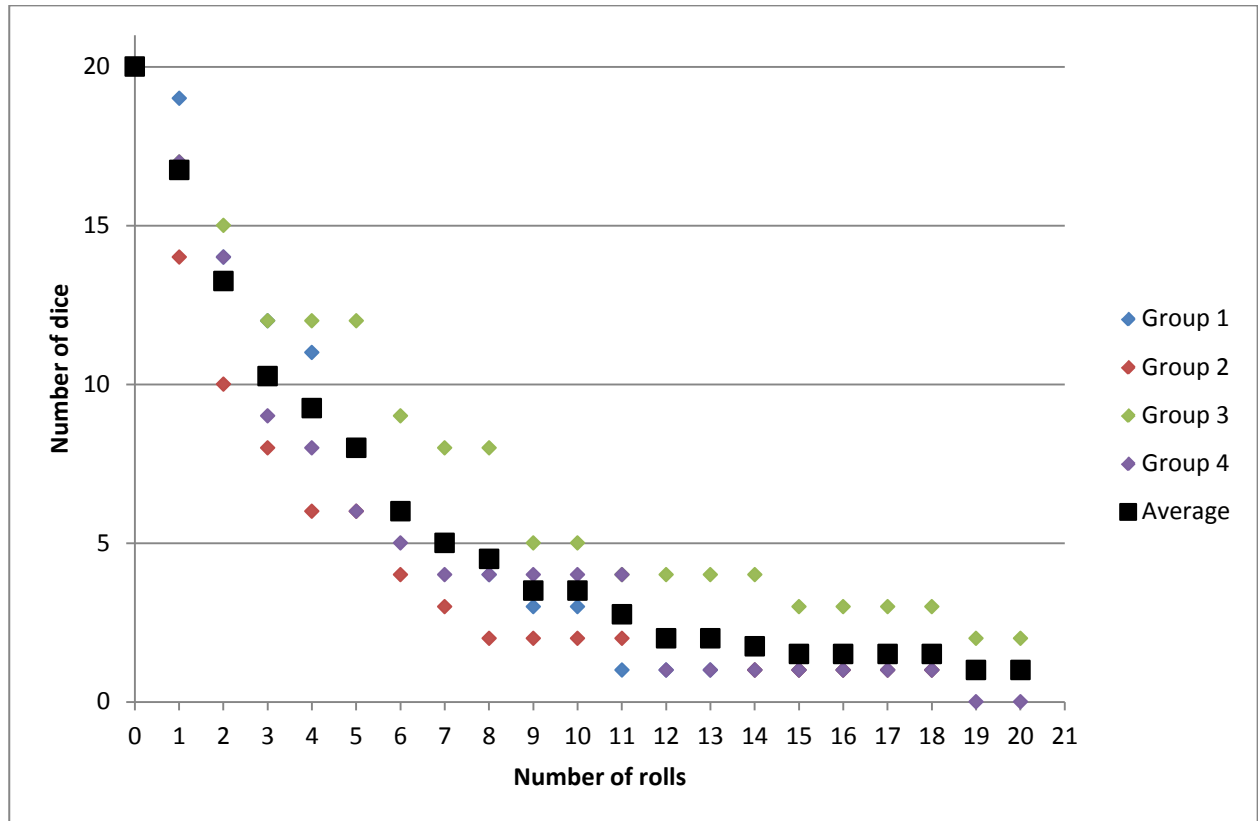
- Provide 100 dice and get the students to follow the instructions on the worksheet.
- Have each group record their own results, and average their results with that of previous groups – this will improve the shape of the decay and give a more accurate half-life. You can explain to them that as the sample size is small and the decay of each individual ‘nucleus’ is random, taking an average of a large sample size gives a better result. This could also be well illustrated by plotting the running total after each roll.
- Ask the students to predict when the ‘marked’ dice will decay - this should demonstrate that the decay of individual nuclei is random, and cannot be predicted, despite the half-life of the material being well defined.

Option 2: Whole class activity:

- You may not have access to multiple sets of 100 dice (!), so can provide 20 or so dice to each group – it is important that each group begins with the same number.
- Ask each group to roll their dice at the same time, and keep a running total of the dice left ‘in play’ in the room. Have each group record their own data and an average from all the groups.
- Plotting both the total and average results should result in two decay curves, which will have different magnitudes, but should both be within a half-life of 3 rolls.
- If you can include one ‘marked’ die in each set you can ask each groups to predict when their marked dice will decay and then plot these incidents on the same graph as the decay curve. This should show that the decay of individual nuclei is random, and cannot be predicted, despite the half-life of the material being well defined.

Example:

Below is an example of the data for 4 groups, starting with 20 dice each. The average data sits closer to the true exponential decay, with the number of dice halving after approximately 3 rolls, as you would expect.



Half-Life Experiment

Introduction:

The rate of decay for a particular radioactive isotope is characterised by its half-life: the time taken for half of the nuclei in a sample to decay. Some examples of radioactive isotopes are given in the table below.

Isotope	Half-life	Used for:
Uranium-238	4.47 billion years	Geological dating
Uranium-235	704 million years	Geological dating
Carbon-14	5730 years	Carbon-dating
Fluorine-18	109.77 minutes	PET imaging
Yttrium-90	2.67 days	Radiotherapy

Despite the half-life of these isotopes being well defined the decay of each individual nuclei is random and it is impossible to predict when it will occur.

Another important property of half-life is that it is the same for any sample size – e.g. half of 1 million atoms of carbon-14 decay in the same time as half of 1 billion atoms of carbon-14.

Activity:

You should have a container with a large number of dice, including 1 'marked' in some way to distinguish it from the others. Each die represents the nucleus of an atom and when a 1 is rolled that nucleus has 'decayed' and is removed from play.

- 1) Make some predictions about the how the 'nuclei' will decay:
 - a. How many rolls do you think you it will take for half of the nuclei to decay? Why?
 - b. How many roles do you think it will take before the marked dice decays? Why?
- 2) Roll the dice, remove all those which show a 1 and record the number of dice remaining. Repeat until no dice remain. Make sure you record the result even if no dice have decayed!
- 3) Combine your data with data from the rest of the class: recording the total and average after each role of the dice.
- 4) Plot the results and draw a line of best fit.

	Small Group Data	Class Data	
Dice Rolls	Number of dice	Average of all groups	Total from all groups
0			
1			
2			
3			
4			
5			
6			
7			
8			
9			
10			
11			
12			
13			
14			
15			
16			
17			
18			
19			
20			
21			
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23			
24			

Use the graphs to answer the following questions:

- 1) Did you correctly predict when the marked nucleus would decay?
- 2) What is the half-life of the dice, measured in numbers of roles? Is this the same for each data set? Did this agree with your prediction?
- 3) Which plot more accurately shows an exponential decay? Why?